

1. THE CRISIS: "NO-BED SYNDROME"

PUBLIC HEALTH CRISIS

In Ghana, traumatic injuries and acute medical emergencies face the severe crisis of **"No-Bed Syndrome"**. Critical ambulances transport unstable patients blindly across metropolitan centers only to be turned away at hospital gates due to 100% ICU/ER saturation. This creates fatal delays during the **Golden Hour** (first 60 mins of trauma).

COMMERCIAL API BARRIER

Google Maps costs \$5-\$10 per '1k' requests, costing thousands monthly for national fleets.

NETWORK DEAD ZONES

Highway transit corridors experience cellular dropouts, crashing traditional thin clients.

2. CLINICAL TRIAGE PROTOCOL (SATS / TEWS)

GHANA HEALTH SERVICE STANDARD

PULSEGRID formalizes triage using the **South African Triage Scale (SATS)** and calculates the composite **Triage Early Warning Score (TEWS)**:

$$TEWS = f(\text{Heart Rate, Systolic BP, Resp Rate, Temp, Mobility, Trauma})$$

SOUTH AFRICAN TRIAGE SCALE (SATS) / TEWS MATRIX

RED (CRITICAL) SATS Score: 1-2 TEWS Score: 1-2	Clinical Criteria: "SATS Score 1-2 OR Physiological Parameters: Systolic BP < 90mmHg, HR > 120bpm, RR > 30bpm, SpO2 < 90%, GCS < 8. Trauma: Significant torso/limb trauma, unstable vitals, or life-threatening injuries." (SATS 1-2 OR 2+ physiological/trauma criteria)
ORANGE (URGENT) SATS Score: 3-4 TEWS Score: 3-4	Clinical Criteria: "SATS Score 3-4 OR Major Trauma: Significant trauma, unstable vitals, or life-threatening injuries." (SATS 3-4 OR major trauma)
YELLOW (STABLE) SATS Score: 5-6 TEWS Score: 5-6	Clinical Criteria: "SATS Score 5-6 OR Minor Trauma: Minor trauma, stable vitals, and no life-threatening injuries." (SATS 5-6 OR minor trauma)
GREEN (MINOR) SATS Score: 7-8 TEWS Score: 7-8	Clinical Criteria: "SATS Score 7-8 OR Minor Injury: Minor injury, stable vitals, and no life-threatening injuries." (SATS 7-8 OR minor injury)
BLUE (MINIMAL) SATS Score: 9-10 TEWS Score: 9-10	Clinical Criteria: "SATS Score 9-10 OR Minor Injury: Minor injury, stable vitals, and no life-threatening injuries." (SATS 9-10 OR minor injury)

Fig. 1: Clinical Triage Matrix & Multi-Parameter TEWS Scoring Protocol

3. SPATIAL GEOMETRIES (POSTGIS)

2,500+ FACILITIES MAPPED

Utilizes **PostgreSQL 16 + PostGIS** spatial indexing (GIST) to ingest all 2,500+ healthcare facilities across all 16 Ghanaian regions from the verified **UN OCHA / HDX HOT-OSM** census. Fast geodesic spatial queries (**ST_DWithin**, **ST_DistanceSphere**) filter candidate facilities in <5ms.

4. FIVE-TIER SYSTEM ARCHITECTURE

DISTRIBUTED MICROSERVICES

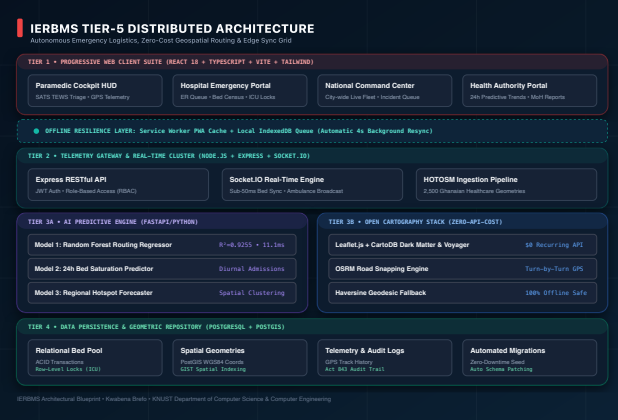


Fig. 3: 5-Tier Pipeline: PMA Edge • Gateway • PostGIS • FastAPI ML • OSRM Engine

5. ZERO-COST OPEN GEOSPATIAL ROUTING

100% FREE & SELF-HOSTED

EVALUATION PARAMETER	COMMERCIAL STACK (GOOGLE MAPS)	PULSEGRID OPEN GEOSPATIAL STACK
Cost per 100k Queries	\$500.00 – \$1,000.00 USD / mo	\$0.00 USD (Self-hosted & Open)
Geographic Dataset	Proprietary Google Network	OpenStreetMap & HOT-OSM Ghana (2,500 facilities)
Routing Machine	Cloud-dependent API	OSRM Road Snapping & Haversine Vector Fallback
Tactical Map Tiles	Raster Standard Google View	CartoDB Dark Matter, Esri Satellite, Street Voyager

6. PARAMEDIC HUD & 3D BED CAPACITY

MISSION-FOCUSED UX



Fig. 4a: Paramedic Cockpit (Turn-by-turn road snapping)

Fig. 4b: Hospital ER 3D Mesh (Pre-arrival bed allocation)

Tactical Mission Isolation: When an ambulance unit accepts a critical emergency, all 2,499 unrelated facility markers vanish from the tactical map. Only the driver's current position, road-snapped OSRM navigation path, and destination hospital are rendered—eliminating driver cognitive overload.

RELATIONAL ENTITY-RELATIONSHIP SCHEMA (POSTGRESQL)

Atomic Bed Allocations, Geospatial Coordinates & Audit Traceability

TABLE: HOSPITALS

PK id : uuid
name : varchar(255)
longitude : double precision
total_general_beds : int4
occupied_general_beds : int4
total_icu_beds : int4
occupied_icu_beds : int4
region : varchar(100)
phone : varchar(50)

TABLE: USERS (RBAC)

PK id : uuid
email : varchar(255) [UNIQUE]
password_hash : varchar(255)
role : enum (paramedic, doc, nurse, admin)
full_name : varchar(255)
phone : varchar(50)
approval_status : varchar(50)
FK hospital_id : uuid
created_at : timestamp

TABLE: HOSPITAL_EQUIPMENT

PK id : uuid
FK hospital_id : uuid
equipment_type : varchar(100)
is_available : boolean
updated_at : timestamp

TABLE: AMBULANCES

PK id : uuid
call_sign : varchar(50)
plate_number : varchar(50)
status : enum (avail, en-route, arrived)
current_latitude : double
current_longitude : double
speed : float4
heading : int4 (0-360 deg)
last_ping : timestamp

TABLE: EMERGENCY_CASES (CLINICAL TRIAGE & TELEMETRY BRIDGE)

PK id : uuid
FK ambulance_id : uuid
FK assigned_hospital_id : uuid
patient_identifier : varchar(100) [TOKENIZED]
trauma_level : int4 (SATS TEWS 0-10)
emergency_type : varchar(100)
bed_type_assigned : varchar(50) (icu, general, trauma)
status : enum (active, in-transit, arrived, resolved)
patient_vitals : jsonb { HR, BP, SpO2, Temp, RR, AVPU }
triage_notes : text
ai_recommended_hospital_id : uuid
ai_estimated_duration_mins : float4
created_at : timestamp
handover_timestamp : timestamp
resolved_at : timestamp

IERBMS Schema Specification • Implemented in PostgreSQL + PostGIS with ACID Row-Level Concurrency Locks

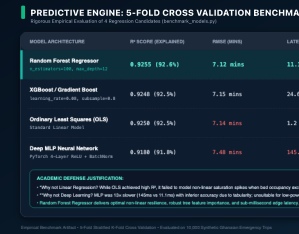
Fig. 2: Relational PostGIS Spatial Schema & Telemetry Data Model

7. MACHINE LEARNING HOSPITAL MATCHING

RANDOM FOREST REGRESSOR

Trained on historical hospital intake patterns. Enforces **Hard Clinical Guardrails** (distance ≤ 60 km, specialized ICU bed > 0 for trauma ≥ 4) + predicts total turnaround latency:

MODEL ARCHITECTURE	R ² SCORE	RMSE (MINS)	MAE (MINS)	LATENCY
Ordinary Least Squares (OLS)	0.9250	7.14	5.68	1.2 μs
Decision Tree (CART)	0.8672	9.48	7.44	3.5 μs
Multi-Layer Perceptron (DNN)	0.9180	7.42	5.92	142.0 μs
Random Forest (PULSEGRID)	0.9255	7.12	5.67	11.1 μs

Fig. 5: 5-Fold Cross-Validation: R² Fit vs Inference Latency

8. TWO-TIER OFFLINE SYNC ENGINE

DEAD-ZONE FAULT TOLERANT

When transit units enter cellular dead-zones, vitals and triage status updates are intercepted by the **Service Worker** and persisted into an **IndexedDB Local Storage Queue**. The client listens for network recovery (navigator.onLine) and flushes the queue via transactional HTTP payloads to PostgreSQL without data loss.

9. QUANTITATIVE CLINICAL IMPACT

SIMULATED BENCHMARK

-35% AMBULANCE TURNAROUND	+28% ICU UTILIZATION	2,500+ FACILITIES INGESTED	\$0.00 RECURRING API FEES
-------------------------------------	--------------------------------	--------------------------------------	-------------------------------------

10. LIVE DEPLOYMENT & ACT 843

SCAN FOR LIVE DEMO



SCAN WITH PHONE CAMERA

<https://final-year-project-frontend-fawn.vercel.app/>
Instant interactive test of Paramedic Cockpit, Hospital ER Triage, and National Telemetry Grid across Ghana.

Regulatory Compliance: Formulated in compliance with the **Ghana Data Protection Act, 2012 (Act 843)**. Enforces Role-Based Access Control (RBAC), end-to-end tokenized sessions, and clinical data minimization.